Column Design for Axial Load and Biaxial Bending

The IS-456 Rule: Minimum Eccentricity (e_x and e_y)

In the real world, a column is never perfectly straight, and loads are never perfectly centered. The IS 456:2000 code mandates that every column must be designed to handle a minimum "accidental" eccentricity.

e_x (Eccentricity in X-direction): $\boldsymbol{e_x = \frac{L}{500} + \frac{D}{30}}$

e_y (Eccentricity in Y-direction): $\boldsymbol{e_y = \frac{L}{500} + \frac{b}{30}}$

(Where L is the unsupported length of the column, D is the depth, and b is the width).

The Biaxial Moments M_x & M_y

- The moment about the X-axis (M_x) is created because the load is offset along the Y-axis.
  i.e, $\mathbf{M_x = P \times e_y}$
- The moment about the Y-axis (M_y) is created because the load is offset along the X-axis.
  i.e, $\mathbf{M_y = P \times e_x}$

(This means you are no longer designing simple axially loaded columns; you are designing columns subjected to biaxial bending (bending in both directions).)

Final design moments for Column (A1)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u=49.06KN

Frame Moments: Mx = 9.32 KN-m**, My =** 5.89 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt **e_x = 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt **e_y = 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

M_x (accidental) = P × e_y = 49.06 KN × 0.02 m = **0.98 KN-m**

M_y (accidental) = P × e_x = 49.06 KN × 0.02 m = **0.98 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design M_x: Max of (9.32 KN-m from frame AND and 0.98 KN-m accidental) = **9.32 KN-m**

Design M_y: Max of (5.89 KN-m from frame AND and 0.98 KN-m accidental) = **5.89 KN-m**

Final SP-16 parameters for Column A1 are:

1. **P =** 49.06 KN
2. **Mx =** 9.32 KN-m
3. **My =** 5.89 KN-m

Final design moments for Column (B1)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 88.69KN

Frame Moments: Mx = 9.81 KN-m**, My =** 1.73 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 88.69 KN × 0.02 m = **1.77 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 88.69 KN × 0.02 m = **1.77 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design M_x: Max of (9.81 KN-m from frame AND and 1.77 KN-m accidental) = **9.81 KN-m**

Design M_y: Max of (1.73 KN-m from frame AND and 1.77 KN-m accidental) = **1.77 KN-m**

Kani's Method analysis said the horizontal bending moment was only 1.73 KN-m. But the IS-456 code says that because of accidental real-world shifting, the column will actually experience at least 1.77 KN-m of bending.

Final SP-16 parameters for Column B1 are:

1. **P =** 88.69 KN
2. **Mx =** 9.81 KN-m
3. **My =** 1.77 KN-m

Final design moments for Column (C1)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 53.94KN

Frame Moments: Mx = 9.73 KN-m**, My =** 8.04 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt **e_x = 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt **e_y = 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

M_x (accidental) = P × e_y = 53.94 KN × 0.02 m = **1.08 KN-m**

M_y (accidental) = P × e_x = 53.94 KN × 0.02 m = **1.08 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design M_x: Max of (9.73 KN-m from frame AND 1.08 KN-m accidental) = **9.73 KN-m**

Design M_y: Max of (8.04 KN-m from frame AND 1.08 KN-m accidental) = **8.04 KN-m**

Final SP-16 parameters for Column C1 are:

1. **P =** 53.94 KN
2. **Mx =** 9.73 KN-m
3. **My =** 8.04 KN-m

Final design moments for Column (A2)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 109.43KN

Frame Moments: Mx = 2.48 KN-m**, My =** 2.06 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 109.43 KN × 0.02 m = **2.19 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 109.43 KN × 0.02 m = **2.19 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design $\mathbf{M_x}$: Max of (2.48 KN-m from frame AND 2.19 KN-m accidental) = **2.48 KN-m**

Design $\mathbf{M_y}$: Max of (2.06 KN-m from frame AND 2.19 KN-m accidental) = **2.19 KN-m**

Kani's Method analysis said the horizontal bending moment was only 2.06 KN-m. But the IS-456 code says that because of accidental real-world shifting, the column will actually experience at least 2.19 KN-m of bending.

Final SP-16 parameters for Column A2 are:

1. **P =** 109.43 KN
2. **Mx =** 2.48 KN-m
3. **My =** 2.19 KN-m

Final design moments for Column (B2)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 114.67 KN

Frame Moments: Mx = 2.97 KN-m**, My =** 3.58 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 114.67 KN × 0.02 m = **2.29 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 114.67 KN × 0.02 m = **2.29 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design $\mathbf{M_x}$: Max of (2.97 KN-m from frame AND 2.29 KN-m accidental) = **2.97 KN-m**

Design $\mathbf{M_y}$: Max of (3.58 KN-m from frame AND 2.29 KN-m accidental) = **3.58 KN-m**

Final SP-16 parameters for Column B2 are:

1. **P =** 114.67 KN
2. **Mx =** 2.97 KN-m
3. **My =** 3.58 KN-m

Final design moments for Column (C2)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 83.94 KN

Frame Moments: Mx = 0.41 KN-m**, My =** 7.31 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 83.94 KN × 0.02 m = **1.68 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 83.94 KN × 0.02 m = **1.68 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design M_x: Max of (0.41 KN-m from frame AND 1.68 KN-m accidental) = **1.68 KN-m**

Design M_y: Max of (7.31 KN-m from frame AND 1.68 KN-m accidental) = **7.31 KN-m**

Kani's Method analysis yielded a very small vertical bending moment (0.41 KN-m) for this specific joint. However, because of the 83.94 KN load sitting on top of it, the IS-456 code requires you to design for a minimum of 1.68 KN-m to stay safe.

Final SP-16 parameters for Column C2 are:

1. **P =** 83.94 KN
2. **Mx =** 1.68 KN-m
3. **My =** 7.31 KN-m

Final design moments for Column (A3)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 76.20 KN

Frame Moments: Mx = 12.18 KN-m**, My =** 0.41 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 76.20 KN × 0.02 m = **1.52 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 76.20 KN × 0.02 m = **1.52 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design M_x: Max of (12.18 KN-m from frame AND 1.52 KN-m accidental) = **12.18 KN-m**

Design M_y: Max of (0.41 KN-m from frame AND 1.52 KN-m accidental) = **1.52 KN-m**

Kani's method analysis showed only 0.41 KN-m of bending from the horizontal frame, but the code requires you to design for at least 1.52 KN-m to account for accidental load shifting. **The accidental moment governs the Y-axis design.**

Final SP-16 parameters for Column A3 are:

1. **P =** 76.20 KN
2. **Mx =** 12.18 KN-m
3. **My =** 1.52 KN-m

Final design moments for Column (B3)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 75.75 KN

Frame Moments: Mx = 6.32 KN-m**, My =** 4.88 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 75.75 KN × 0.02 m = **1.52 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 75.75 KN × 0.02 m = **1.52 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design $\mathbf{M_x}$: Max of (6.32 KN-m from frame AND 1.52 KN-m accidental) = **6.32 KN-m**

Design $\mathbf{M_y}$: Max of (4.88 KN-m from frame AND 1.52 KN-m accidental) = **4.88 KN-m**

T he frame moments from your Kani's Method analysis were strong enough to safely overpower the IS-456 minimum eccentricity rules. **The frame moments govern the design in both directions.**

Final SP-16 parameters for Column B3 are:

1. **P =** 75.75 KN
2. **Mx =** 6.32 KN-m
3. **My =** 4.88 KN-m

Final design moments for Column (C3)

From our previous structural analysis and slab designs, we have our starting data locked in:

Dimensions: Width b=230mm, Depth D=300mm

Unsupported Length: L=2700mm

Factored Axial Load: P_u= 52.85 KN

Frame Moments: Mx = 9.22 KN-m**, My =** 7.17 KN-m ***(From FEM map)***

Materials: M20 Concrete (f_{ck}=20N/mm^2)and Fe500 Steel (f_y=500N/mm^2)

Calculation of Minimum Eccentricity

$e_x = \frac{L}{500} + \frac{D}{30} = \frac{2700}{500} + \frac{300}{30}$ = 15.4mm (Since 15.4 mm < 20 mm, adopt $\mathbf{e_x}$ **= 20 mm** or 0.02 m)

$e_y = \frac{L}{500} + \frac{b}{30} = \frac{2700}{500} + \frac{230}{30}$ = 13.06mm (Since 13.06 mm < 20 mm, adopt $\mathbf{e_y}$ **= 20 mm** or 0.02 m)

Calculate Accidental Eccentric Moments

$\mathbf{M_x}$ **(accidental)** = P × e_y = 52.85 KN × 0.02 m = **1.06 KN-m**

$\mathbf{M_y}$ **(accidental)** = P × e_x = 52.85 KN × 0.02 m = **1.06 KN-m**

Final Design Moments

(Take the maximum between the Frame Moment and the Accidental Moment)

Design $\mathbf{M_x}$: Max of (9.22 KN-m from frame AND 1.06 KN-m accidental) = **9.22 KN-m**

Design $\mathbf{M_y}$: Max of (7.17 KN-m from frame AND 1.06 KN-m accidental) = **7.17 KN-m**

T he frame moments from your Kani's Method analysis were strong enough to safely overpower the IS-456 minimum eccentricity rules. **The frame moments govern the design in both directions.**

Final SP-16 parameters for Column C3 are:

1. **P =** 52.85 KN
2. **Mx =** 9.22 KN-m
3. **My =** 7.17 KN-m